



ELECTRICAL MEASUREMENTS

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ELECTRICAL MEASUREMENTS

Scope of presentation

Sessions Policy

Objectives

Introduction to sensors
and transducers

Sensors
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Quality parameters for
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Thanks & Questions



Menoufia University
FACULTY OF Electronic Eng





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الإحترام
المتبادل



الإختلاف
لا يفسد
للود قضية



المشاركة
الفعالة



البشاشة
والإبتسام



Thanks & Questions



Sensors and Transducers

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□ Chapter Learning Outcomes

❖ **After completing this chapter, the student will be able to:**

- ✓ Describe the performance of commonly used sensors.
- ✓ Describe the sensor classification.
- ✓ Evaluate sensors used in the measurement of motion, position, force, flow and temperature.

Thanks & Questions



Sensors and Transducers

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☐ *Lecture Outline :*

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☐ Introduction to sensors and transducers

☐ Sensors Classifications

☐ Instrumentation System

☐ Quality parameters for Sensors

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Introduction to sensors and transducers

Definition

Sensor is a device that produces an output signal for the purpose of sensing of a physical variable (i.e., pressure , temperature , motion,...).

Transducer is a device that converts a signal from one physical form to a corresponding signal, which has a different physical form.

OR

It is a device that converts a signal from one physical form to a corresponding signal, which has a different physical form.

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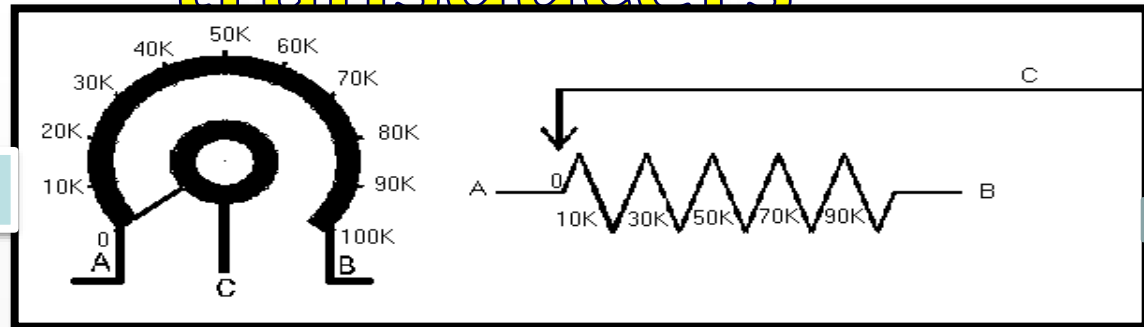
Introduction to sensors and transducers

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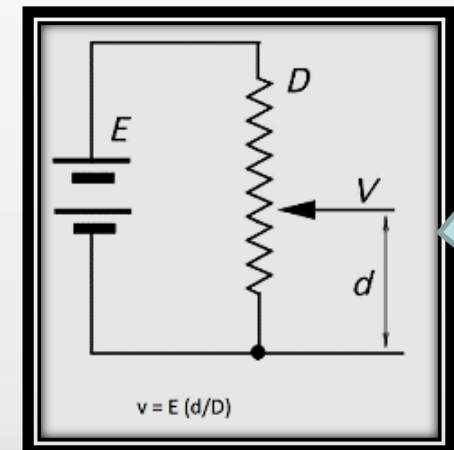
Quality parameters for Sensors

Sensor



Convert the rotary motion into a change in the resistance .

Transducer



Convert the change in the resistance into output voltage

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Introduction to sensors and transducers

Selection of a sensor or a transducer

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Variables measured and application.

The nature of precision and the sensitivity required for the measurement.

Dynamic range.

Level of automation.

Complexity of the control system and modeling requirements.

Cost, size, usage, and ease of maintenance.

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Mechanical

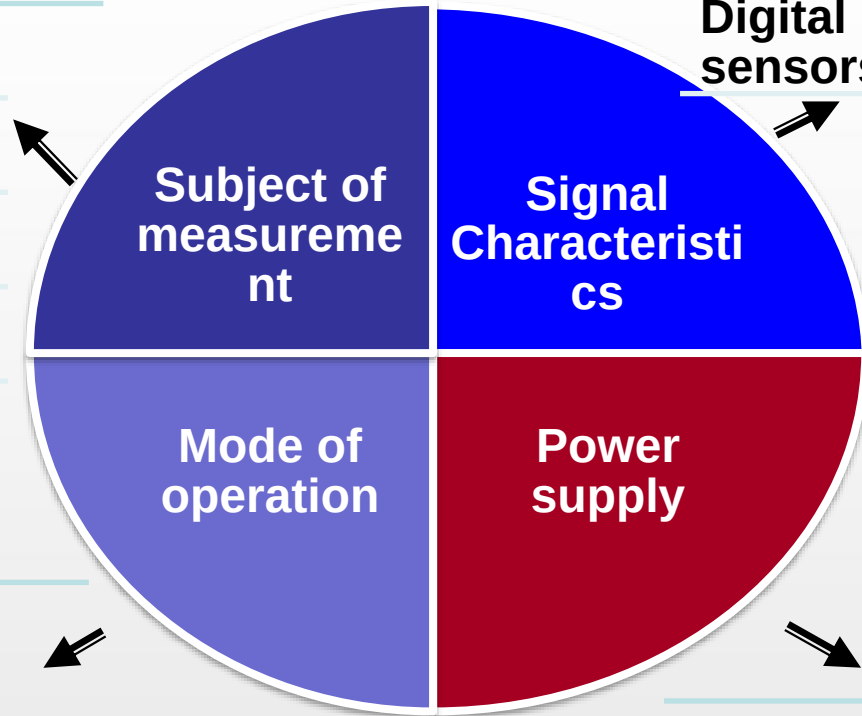
Chemical

Thermal

.....

Null-Type

Deflection-type



Analog
sensors

Digital
sensors

Passive

Active

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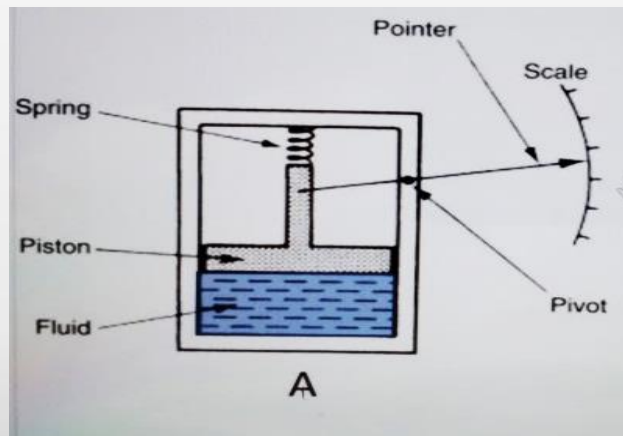
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Analog and digital Sensores

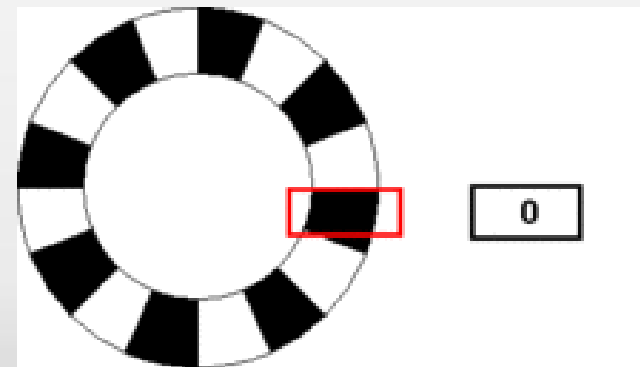
Analog Sensor

It gives output that varies continuously as the measured quantity.



Digital sensor

It gives output that varies in discrete steps and only has finite number of values.



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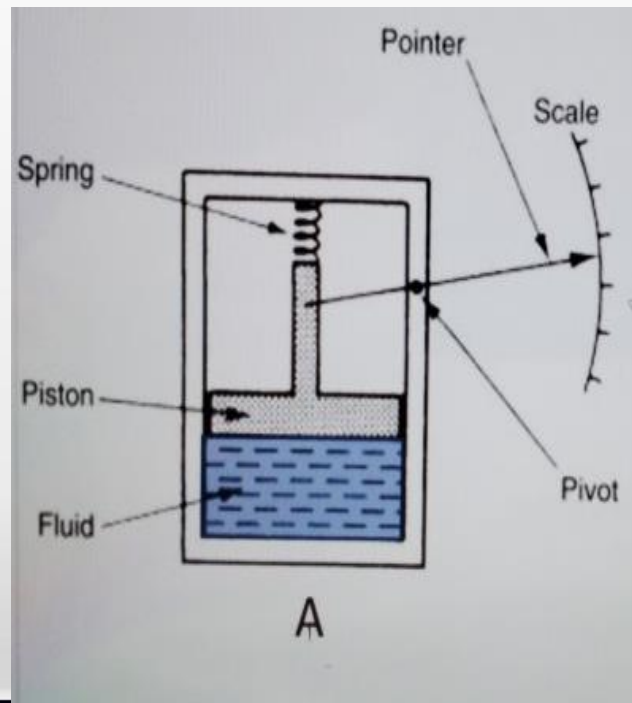
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Passive and Active Sensors

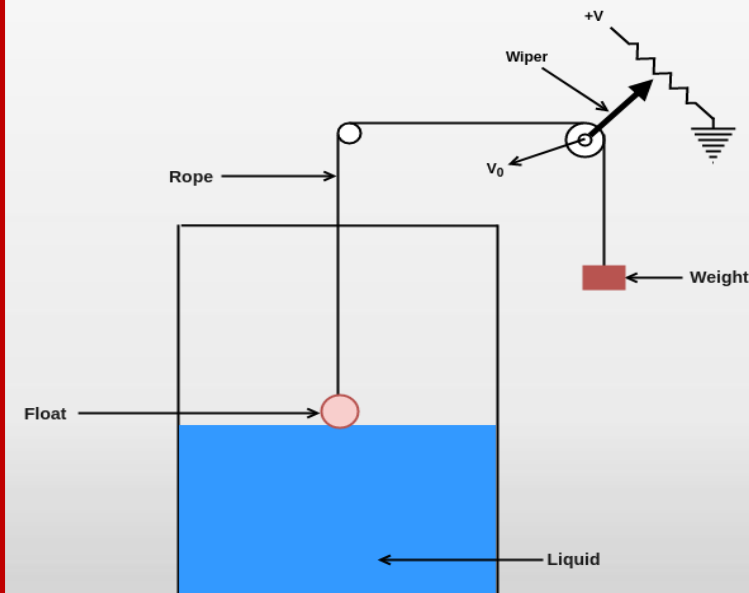
Passive

Whose output is entirely produced by the measured quantity.



Active

The measured value simply modulates the magnitude of some external power source.



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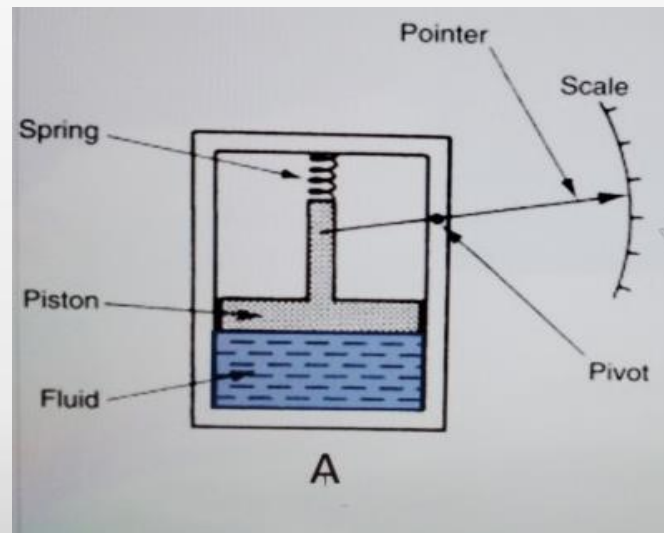
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Deflection and Null-type Sensors

The deflection-type

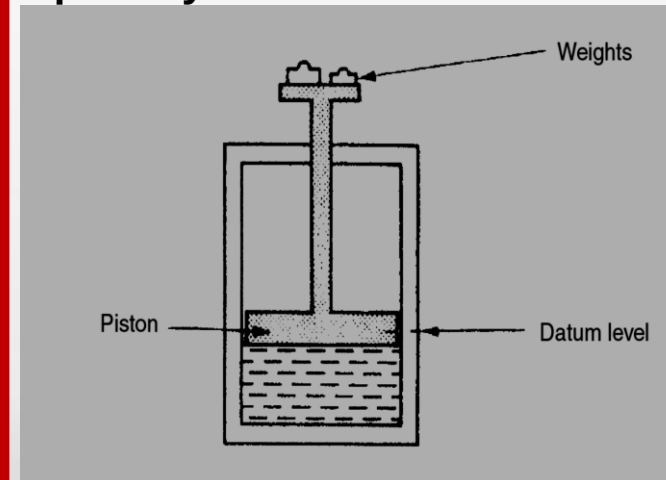
The measured value is displayed in terms of the amount of a deflection movement of a pointer.



Pressure Gauge

The Null-type

In which zero or null indication determines the magnitude of the measured quantity.



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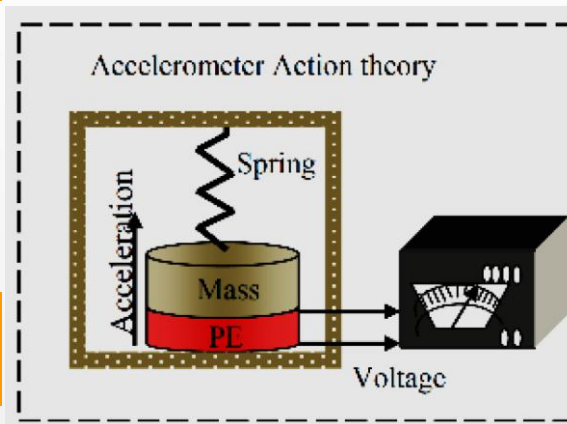
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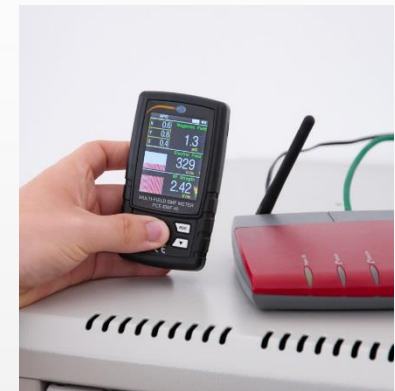
Subject of measurement



Mechanical Sensor



Temperature Sensor



Radiation Sensor

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1. Static Characteristics

- *Range and span*
- *Error*
- *Accuracy*
- *Repeatability & Precision*
- *Resolution*
- *Sensitivity*
- *Hysteresis Error*
- *Non Linearity Error*
- *Stability*
- *Dead Band*
- *Output Impedance*

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2. Dynamic Characteristics

- *Response Time*
- *Time Constant*
- *Rise Time*
- *Settling Time*

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Quality parameters for Sensors

The performance of any sensor is affected by several factors which are:

Sensor Characteristics

Sensitivity

Resolution

Accuracy

Precision

Backlash

Repeatability

Linearity

Rang or span

Tolerance

Threshold

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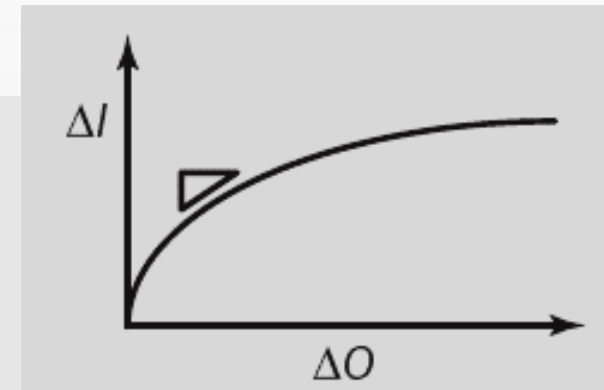
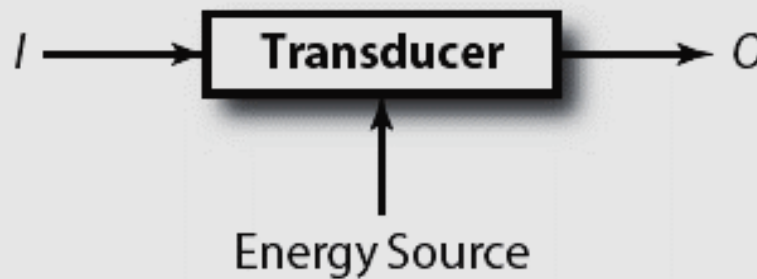
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Sensitivity

It can be expressed as the ratio of change of output to change of input



$$S = \frac{\Delta O}{\Delta I}$$

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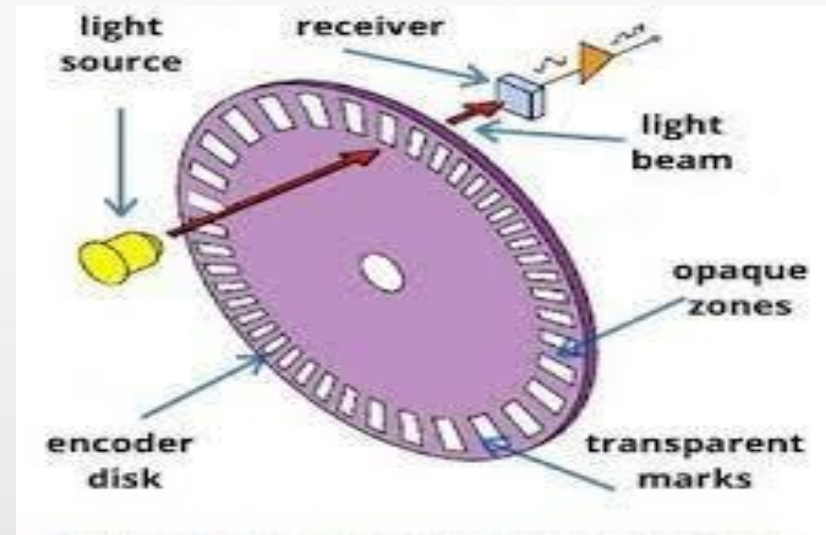
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Resolution

Resolution is defined as the smallest increment in the measured value that can be detected

$$\text{Resolution} = \frac{360}{\text{Number of holes}}$$

Assume Number of holes=20



$$\text{Resolution} = \frac{360}{20} = 18 \text{ degree}$$

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Accuracy

Accuracy is a measure of the difference between the measured value and actual value.

- In practice, the accuracy is defined as a percentage of the true value.

$$\text{percentage of true value} = \frac{\text{measured value} - \text{true value}}{\text{true value}} \times 100\%$$

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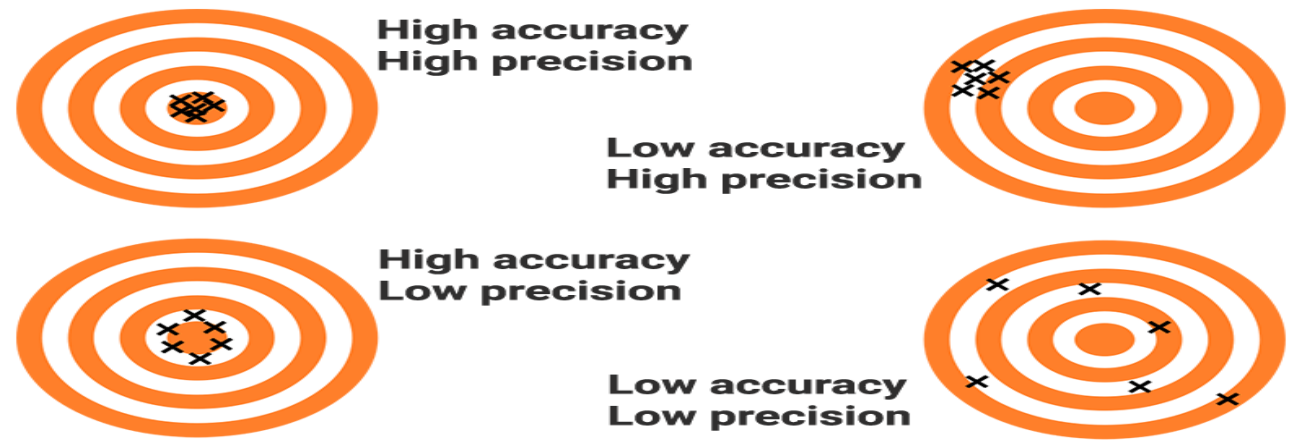
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Precision

Precision is the ability of an instrument to reproduce a certain set of readings within a given accuracy.



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Backlash

Backlash is defined as the maximum distance or angle through which any part of a mechanical system can be moved in one direction without causing any motion of the attached part.

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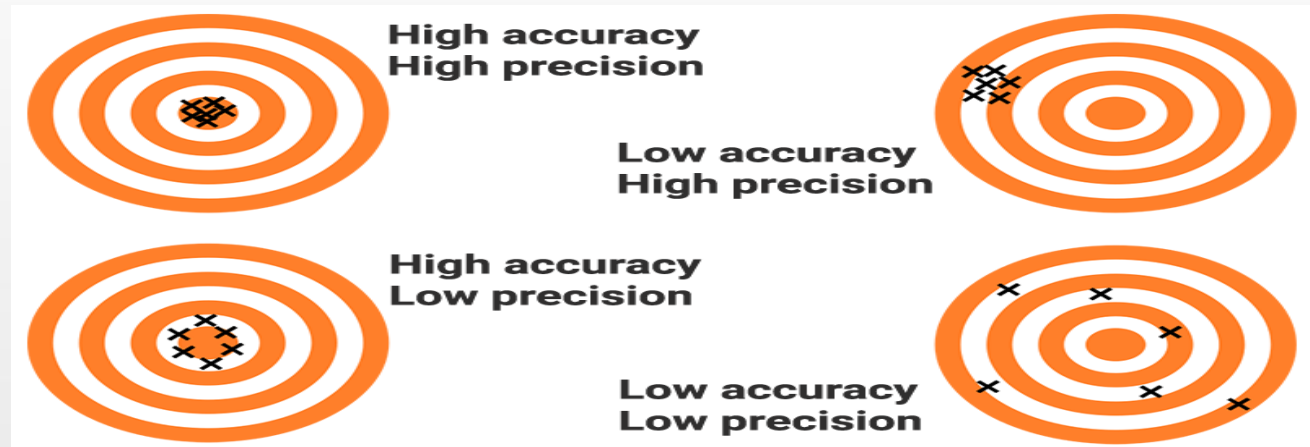
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Repeatability

Repeatability is the ability to reproduce the output signal exactly when the same measurand is applied repeatedly under the same environmental conditions.



$$\% \text{ Repeatability} = \frac{\text{maximum reading} - \text{minimum reading}}{\text{full scale reading}} * 100$$

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- **Reproducibility** describes the closeness of output readings for the same input when there are changes in the method of measurement, observer, measuring instrument, location, conditions of use and time of measurement.
- Both terms thus describe the spread of output readings for the same input.

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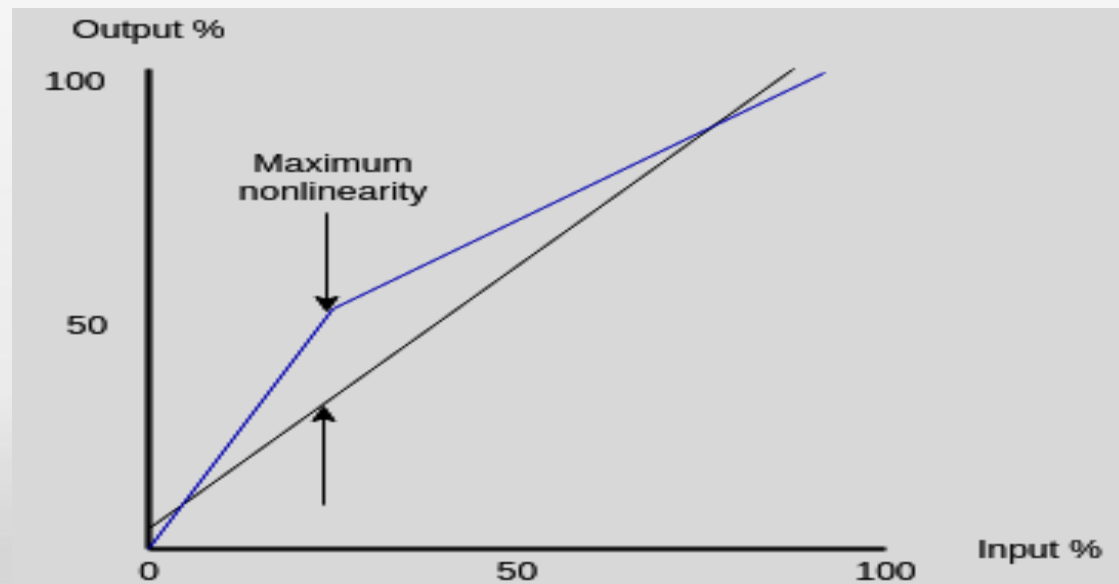
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Linearity

- Ideally, the output reading of an instrument is linearly proportional to the quantity being measured.
- Practically, the output reading of an instrument is not linearly proportional to the quantity being measured.





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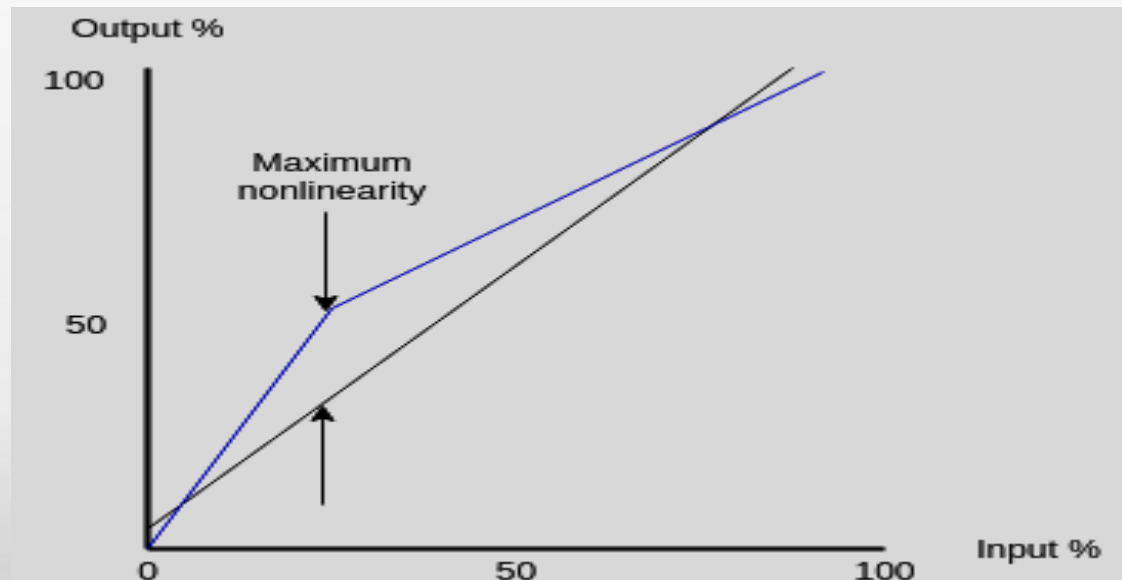
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Linearity

- The linearity is expressed as the percentage of departure from the linear value (i.e., maximum deviation of the output curve from the best-fit straight line during a calibration cycle).



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- If the input to an instrument is gradually increased from zero, the input will have to reach a certain minimum level before the change in the instrument output reading is of a large enough magnitude to be detectable. This minimum level of input is known as the threshold of the instrument.
- When an instrument is showing a particular output reading, there is a lower limit on the magnitude of the change in the input measured quantity that produces an observable change in the instrument output. Like threshold, resolution is sometimes specified as an absolute value and sometimes as a percentage of f.s. deflection.



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- Zero drift or bias describes the effect where the zero reading of an instrument is modified by a change in ambient conditions. This causes a constant error that exists over the full range of measurement of the instrument.
- Sensitivity drift (also known as scale factor drift) defines the amount by which an instrument's sensitivity of measurement varies as ambient conditions change. It is quantified by sensitivity drift coefficients that define how much drift there is for a unit change in each environmental parameter that the instrument characteristics are sensitive to.



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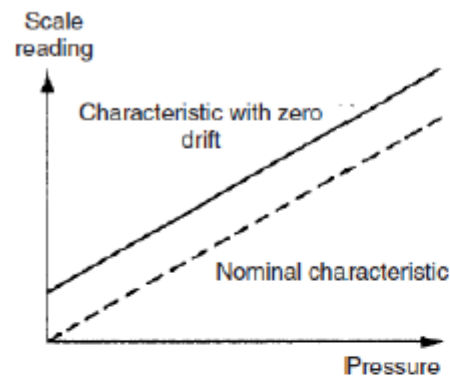
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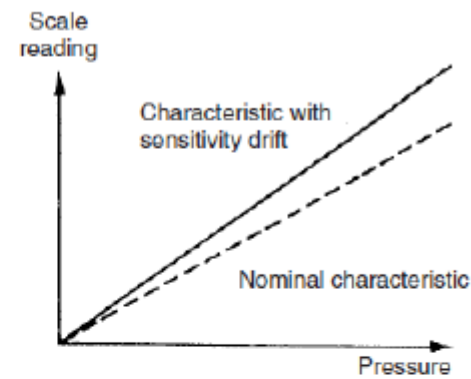
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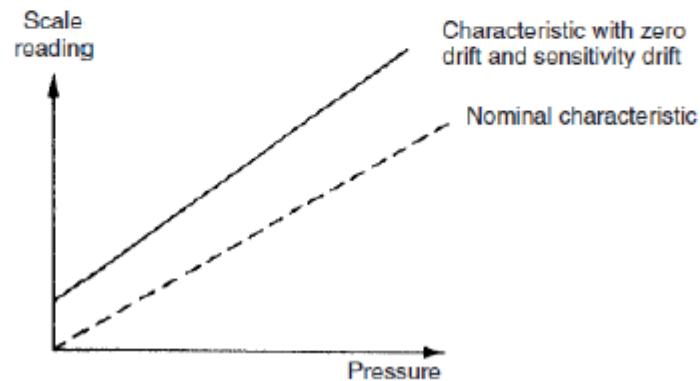
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(a)



(b)



(c)



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- **Tolerance** is a term that is closely related to accuracy and defines the maximum error that is to be expected in some value.
- **The range or span** of an instrument defines the minimum and maximum values of a quantity that the instrument is designed to measure.

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- **Static characteristics** of a transducer are those characteristics when the output signal is in the steady state in response to an input.
- **Dynamic characteristics** of a transducer are those characteristics when the output signal is in the Transient state in response to an input
- Response time (time taken to rise to 95% of the actual value)
- Time constant (Inertia of the sensor, 63.2% of the response time)

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- **Rise time** (It is the time taken by a signal to change from a specified low value to a specified high value. Typically, in analog electronics, these values are 10% and 90% of the step height)
- **Settling time** (time taken to settle within 2% of the steady state value)

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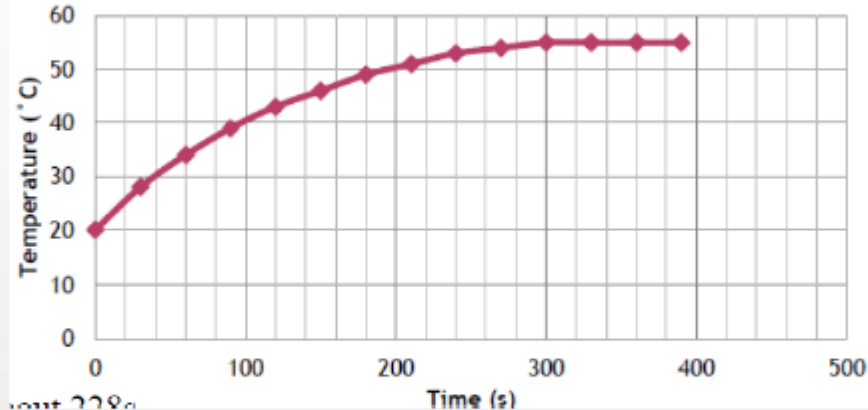


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Time(s)	0	30	60	90	120	150	180	210	240	270	300	330	360	390
Temp. (°C)	20	28	34	39	43	46	49	51	53	54	55	55	55	55

Temperature Vs. Time



Example Steady state value = 55°C 95 % of 55°C is 52.5°C and so the response time about 228s Time constant $\approx$ 60s and Settling time $\approx$ 270s



Questions and Discussion

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End of Lecture

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Good Luck
See You in
Next Lecture . . .



With My Best Wishes
Dr. Eng. Mohamed Abdo